Project Overview (30-Second Version)

What it is: A model that predicts whether a credit card customer will default on their payment next month, trained on the real UCI "Default of Credit Card Clients" dataset — 30,000 customers, ~22% of whom defaulted.

The pipeline, in order:

1. **Clean the data** — fix two columns (`EDUCATION`, `MARRIAGE`) that have undocumented category codes.
2. **Explore the data** — plots in Pandas/Matplotlib/Seaborn to understand what actually separates defaulters from non-defaulters.
3. **Engineer two features** — `CREDIT_UTIL_RATIO` and `NUM_DELAYED_PAYMENTS`, because the EDA shows these matter a lot more than raw dollar amounts.
4. **Train three models** — Logistic Regression, Decision Tree, Random Forest — tune each with `GridSearchCV`, and pick the best by F1-score.
5. **Evaluate properly** — not just accuracy, because the data is imbalanced (22% defaulters). Accuracy, precision, recall, F1, and a confusion matrix.
6. **Deploy it** — a FastAPI REST API with Pydantic validation so bad input never reaches the model.

The one-sentence answer if an interviewer asks "walk me through it":

> *"I took a real imbalanced classification problem — predicting credit card default — cleaned the data, found through EDA that recent payment behavior matters far more than raw balances, engineered two features around that, trained and compared three models with cross-validated hyperparameter tuning, handled the class imbalance with class weights, evaluated with precision/recall/F1 instead of just accuracy, and shipped the best one behind a validated REST API."*

Actual results (your numbers — see `02_models_explained.md` for detail):

Model	Accuracy	Precision	Recall	F1
Logistic Regression	76.3%	0.47	0.56	0.511
Decision Tree	75.5%	0.46	0.59	0.514
Random Forest (winner)	**77.5%**	**0.49**	**0.61**	**0.544**

Read the other files in this folder in order — each one is written so you can explain that part of the project out loud without notes.

- `01_eda_plots_explained.md` — every plot, what it shows, what it means
- `02_models_explained.md` — how each model works, why these three, why RF won

- `03_metrics_explained.md` — accuracy/precision/recall/F1/confusion matrix, with your real numbers
- `04_interview_qa.md` — the questions you're most likely to get, answered